

Multiclass classification - Introduction

A classification problem where there are more than two categories/classes to predict.

Softmax Regression Algorithm

→ Generalization of Logistic Regression

if $a_1 = P(Y \text{ is } 1)$

$a_2 = P(Y \text{ is } 0)$

then,

$$a_1 + a_2 = 1$$

$$\Rightarrow a_1 = 1 - a_2$$

Derivation

if the softmax regression has 4 o/p's, then,

$$z_1 = \vec{w}_1 \cdot \vec{x} + b_1$$

$$\begin{aligned} \text{o/p, } a_1 &= \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3} + e^{z_4}} \\ &= P(Y = 1) \end{aligned}$$

$$a_2 = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3} + e^{z_4}} = P(Y = 2)$$

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$$a_y = \frac{e^{z_y}}{e^{z_1} + e^{z_2} + e^{z_3} + e^{z_4}} = P(Y=y)$$

Softmax Regression for
N possible outputs

$$z_j = \vec{w}_j \cdot \vec{x} + b_j$$

where $j = 1, 2, \dots, N$.

$$a_j = \frac{e^{z_j}}{\sum_{k=1}^N e^{z_k}} = P(Y=j)$$

where, $\{a_1 + a_2 + \dots + a_j = 1\}$

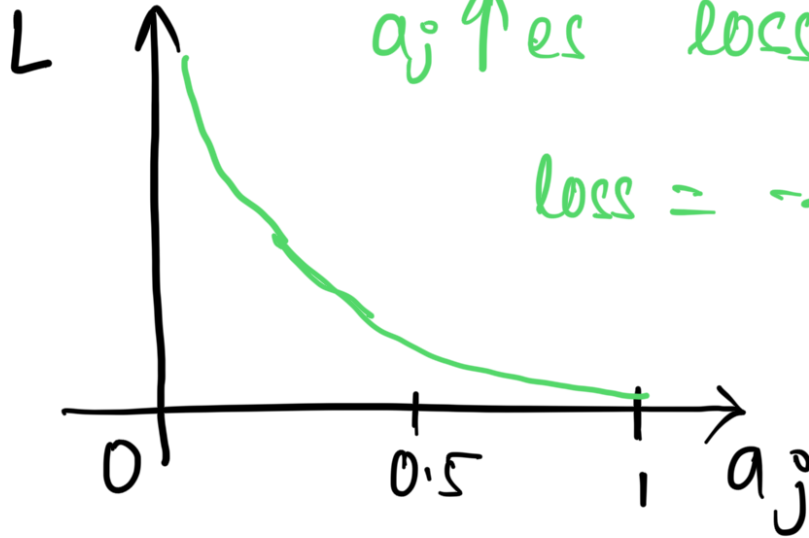
Cost of softmax regression

loss $(a_1, \dots, a_N, y)$

$1 - \log a_y$ if $y \leq 1$

$$= \begin{cases} -\log a_2 & \text{if } y=2 \\ -\log a_N & \text{if } y=N \end{cases}$$

Crossentropy
loss



$a_j \uparrow \Rightarrow \text{loss} \downarrow$

$$\text{loss} = -\log a_j \\ \text{if } y=j$$